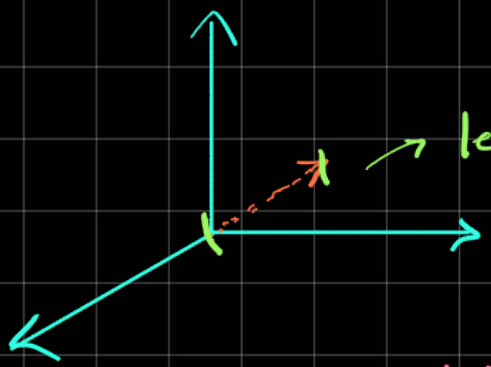
Length of a vector:

$$\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$





steps along x & y axis

$$l^2 = a^2 + b^2$$

coordinates

$$l = \sqrt{x^2 + y^2}$$

$$\begin{bmatrix} 1 \to x \\ 2 \to y \end{bmatrix}$$

Dot product (Inner product)

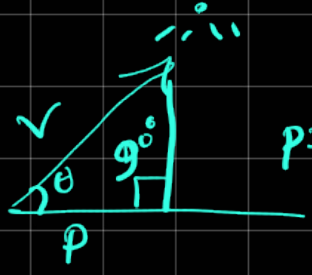


we are multiplying projection lengths.

projection of v on top of u.

proj of u over v.





u

vcosθ

$u \cdot v = |p|\,|u| = v\cos\theta\, u$

$= |u||v|\cos\theta$.

Orthogonality

v_1 is orthogonal ($\perp$) to v_2

$\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \quad \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$ $\quad u \cdot v = 0$

$-2 + 0 + 2 = 0$

$\longrightarrow u^T v \Rightarrow$ why

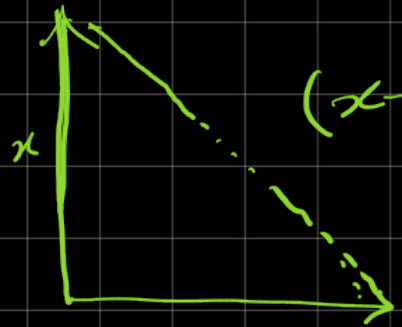
↓

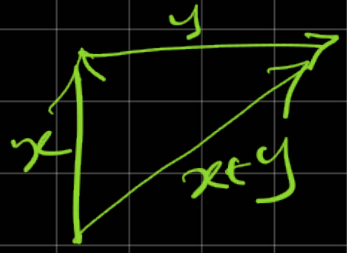
Transposing u casts a shadow over the base vector itself.



$|p| = 0$

$\begin{bmatrix} 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$





$u^T v = 0$ gotta proof.

$$|x|^2 + |y|^2 = |(x+y)|^2$$

$$\Rightarrow (x^T x) + (y^T y) = (x+y)^T (x+y)$$

$$\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \rightarrow 1^2 + 1^2 + 2 = 6$$

$$\begin{bmatrix} 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$1 \cdot 1 + 1 \cdot 1 + 2 \cdot 2 = 6$$

$$(|x|)^2 = x^T x = (x_1\ x_2 \cdots x_3) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_3 \end{pmatrix}$$

$$|x| = \sqrt{x_1^2 + x_2^2 + \cdots x_n^2} = \sqrt{x^T x}$$

$$x^T x + y^T y = (x^T + y^T)(x+y)$$

$x^T x + y^T y = x^T x + x^T y + y^T x + y^T y$

$x^T y + y^T x = 0$

$\Rightarrow 2x^T y = 0$

$x^T y = 0$

$x^T y = y^T x \longrightarrow$ As both vectors.

$[x^T] \begin{bmatrix} y \end{bmatrix}$

$[y^T] \begin{bmatrix} x \end{bmatrix}$ } same multiplication, same result

$\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$

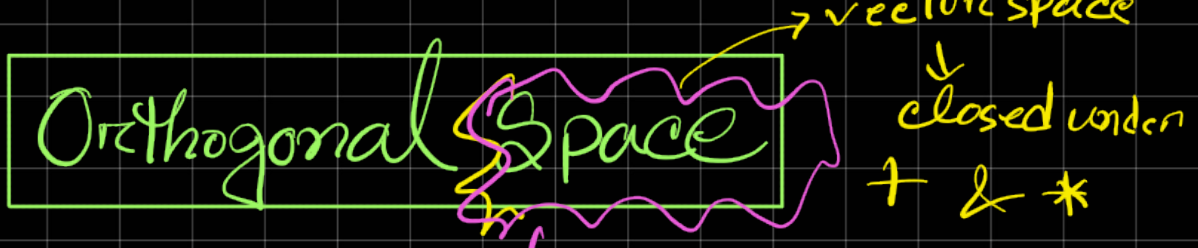
$[1 \ 1 \ 2] \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} = 2 + 2 + 6 = 0$

$[2 \ 2 \ 3] \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

0 vector always orthogonal to every other vector.

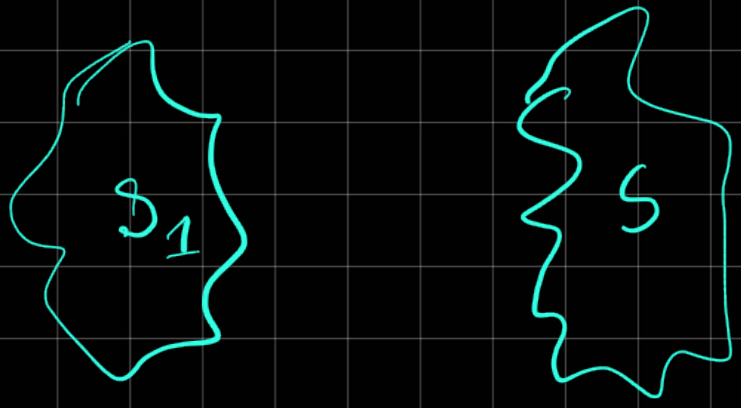
Dot product of any vector to 0

$= 0$



col, row, null, left null.

$C(A), C(A^T), N(A), N(A^T)$



all vectors in S_1 space must be individually orthogonal to all the vectors of space S

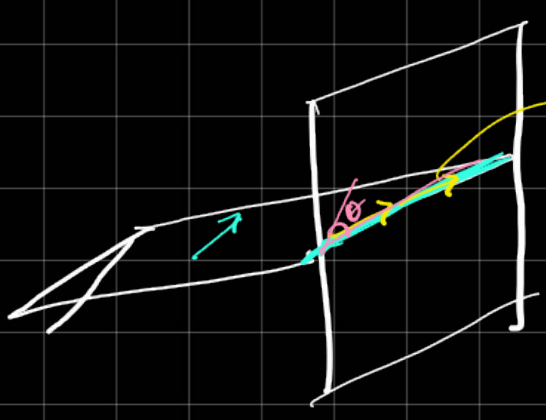
the vectors of space 2.

floor + wall → line. (vectors belong to both space & on top of another not orthogonal.

if

$S_1 \cap S_2 = \{0\}$ → then S_1 & S_2 can be orthogonal.

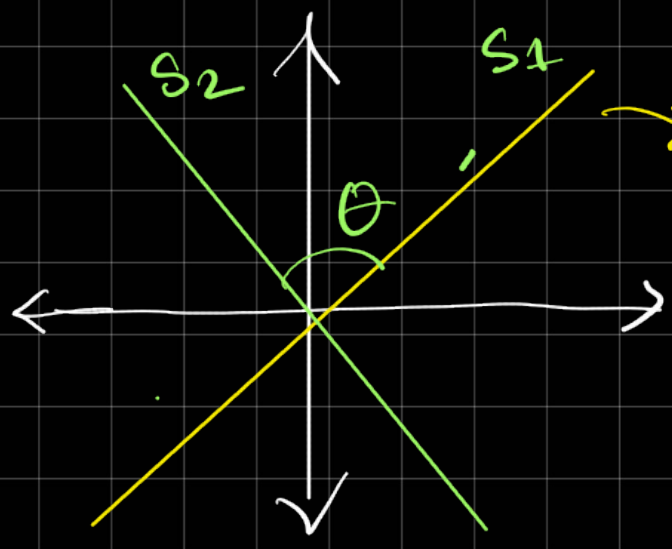
orthogonal complement.



not perpend. on top of another

↓

if only intersected the 0, the advance.



→ not perpendicular.

as vectors of the line belonging to both the spaces lie on top of each other.

(if $\theta = 90°$, S_1 & S_2 are orthogonal complement)

S_1 & S_2 part of same plane.

2 planes can be orthogonal.

Rowspace & Nullspace

$Ax = 0$ → think as x^T → $x^T y = 0$ → part of nullspace

$$\begin{bmatrix} \text{row } 1 \\ \text{row } 2 \\ \vdots \\ \text{row } m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$m \times n$ $n \times 1$ $m \times 1$

$(\text{row}) \cdot [x] = 0$

↑ dot product

rows are perpendicular to vectors x.

So these 2 are orthogonal space.

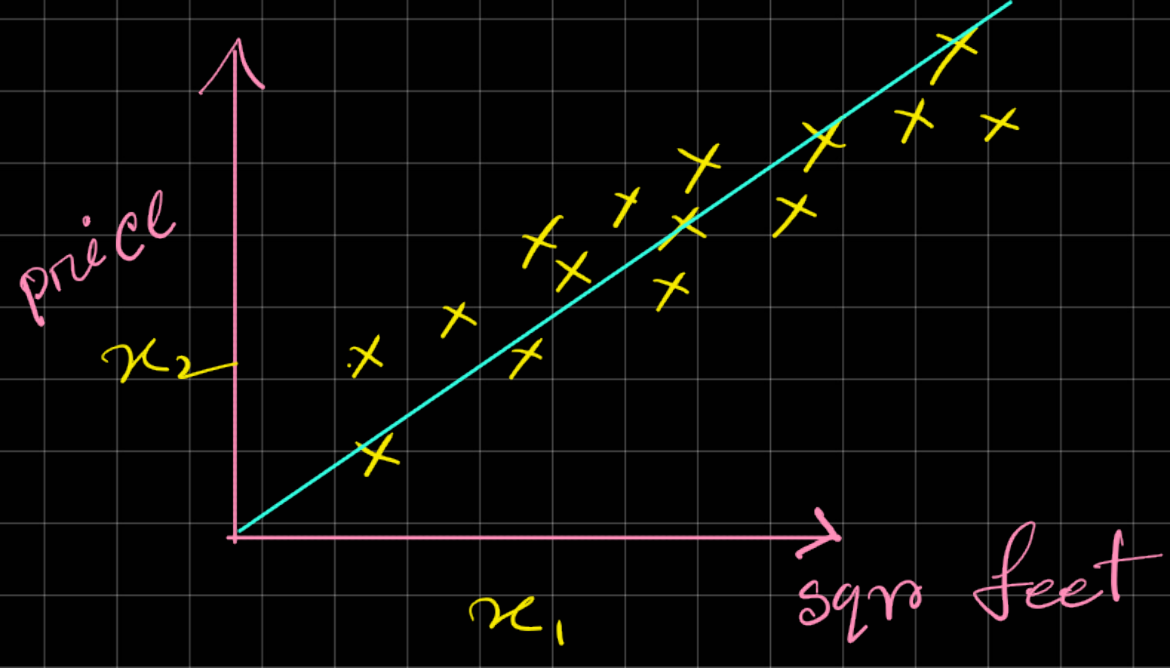
rowspace & Nullspace

colspace & Leftnullspace

orthogonal complement.

Buy a House.

linear trend.



data always noisy, never fall within a single line.

2 variables → 13 different eqn's.

$$\begin{bmatrix} & \\ & \\ & \end{bmatrix}_{13\times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

LA tries to solve a system where there is no soln.

$Ax = b$ → not in colspace of a → no soln

$m > n$ → 1 soln or 0 soln.

=> orthogonality helps.

$$Ax = b$$

$$\underbrace{A^TA}_{n \times n} x = \underbrace{A^Tb}_{n \times 1}$$

A not invertable → A^T invertable.

$$\begin{bmatrix} & A & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

↳ in Machine learning is topics

H.W

$$\begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 3 \\ 3 \\ 2 \end{bmatrix}$$

x_1 x_2

↳ basis for vector space

S. → Find s.

↓

which perpendicular

to x_1 & x_2.

